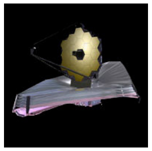

2410 - How efficiently do galaxies produce ionizing photons in the epoch of reionization?

Cycle: 1, Proposal Category: AR

INVESTIGATORS

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OBSERVATIONS

ABSTRACT

Determining the role of galaxies in reionizing the Universe is a fundamental goal in galaxy evolution and cosmology. One of the key variables in reionization models is the ionizing photon production efficiency, which currently remain largely unconstrained at the epoch of reionization. Leveraging datasets from two archival ERS programs (program 1324 and 1345), we propose to investigate the ionizing photon production efficiency of both bright and faint galaxy populations in the epoch of reionization. We will directly constrain ionizing photon production efficiency via measurement of extinction corrected H-alpha luminosity and rest-frame UV continuum luminosity. This will provide, for the first time, the direct constraint on ionizing photon production efficiency at z>6, informing the ionizing photon budgets of galaxies in the reionization epoch. Additionally, we will measure how ionizing photon production efficiency evolves over cosmic time, and how it varies with galaxy properties. Together, these results will serve as crucial inputs in theoretical models mapping the timeline and topology of reionization.

OBSERVING DESCRIPTION

This archival proposal will leverage datasets from two JWST/ERS programs (#1324 and #1345) to investigate the ionizing photon production efficiency of galaxies in the epoch of reionization.

The combine datasets will allow us to measure extinction corrected H-alpha luminosity which will then be compared with rest-frame UV luminosty to estimate the ionizing photon production rate, for the first time, at the epoch of reionization. The two ERS programs are sensitive to two different luminosty regime (bright and faint), and we will investigate how ionizing photon production rate of reionization epoch galaxies differ with luminosity. Furthermore, we will divide our sample into bins of redshit which will allow us to examine evolution (or lack of) of ionizing photon producing efficieny over cosmic time. Finally, we will investigate physical properties of galaxies that are most efficient producer of ionizing photons which will give us clue to sources of cosmic reionization.